The Latent Truth: Investigating the Knowledge-Behavior Gap in Logical Deduction via Activation Steering

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Abstract

Large Language Models (LLMs) often generate logically invalid outputs despite possessing sufficient in-context premises. We formalize this discrepancy as the **Knowledge-Behavior Gap**. Evaluating `gemma-2-2b-it` and `phi-2` (2.7B) across 250 synthetic logic clusters, we identify a sharp, cross-architectural latent phase transition: linear probes decode logical truth with $\sim 100\%$ accuracy by Layer 8, yet behavioral accuracy in Chain-of-Thought and Solver-Augmented regimes plateaus at $\sim 65\%$. We identify a *Solver-Augmented Paradox*: providing explicit ground-truth symbolic logic fails to close this gap. By applying directional activation steering at saturation layers, we show that a calibrated nudge ($\alpha = 1.0$) toward the latent "truth vector" recovers 76.7% (Gemma) and 75.0% (Phi-2) of failures without degrading coherence. Control interventions using random vectors yield 0% recovery, confirming the causal specificity of this logical manifold. Our results suggest logical failures in mid-scale LLMs are not representational deficits, but consistent retrieval failures during auto-regressive decoding, indicating a fundamental bottleneck in final-layer calibration that cannot be resolved through prompt engineering alone.

1 Introduction

As Large Language Models (LLMs) are deployed in autonomous reasoning, understanding their internal logical fidelity is paramount. A critical failure mode is the "hallucination of falsehood," where models generate logically invalid conclusions despite correct in-context premises.

We hypothesize this stems from a **Knowledge-Behavior Gap**: a structural misalignment between latent logical representations and the auto-regressive decoding process. This gap persists even with techniques like Chain-of-Thought (CoT) prompting (Wei et al., 2022), which can be unfaithful to the model's true internal state (Turpin et al., 2024). Applying mechanistic interpretability to `gemma-2-2b-it` and `phi-2` (2.7B), we demonstrate that these models develop near-perfect latent representations of logical truth long before token generation.

Our main contributions are:

- **Empirical Validation:** We show linear probes decode logical truth with $\geq 94.0\%$ accuracy by Layer 8 in both models, reaching 100% saturation by Layer 16, despite behavioral plateaus of $\sim 65\%$.
- **Causal Intervention:** We isolate a "Truth Vector" and prove Activation Steering ($\alpha = 1.0$) recovers 76.7% (Gemma) and 75.0% (Phi-2) of failures; crucially, random vector injections yield 0% recovery, confirming causal specificity.
- **Solver-Augmented Paradox:** We demonstrate that providing explicit, ground-truth symbolic reasoning traces fails to close the Knowledge-Behavior Gap, pinpointing the final-layer unembedding process as the primary reasoning bottleneck.

2 Related Work

Our approach synthesizes advancements in mechanistic interpretability, representation engineering, and the study of unfaithful reasoning.

Activation Steering and Localization. While earlier work established methods for locating and editing localized factual associations within model weights (Meng et al., 2022), recent paradigms like Representation Engineering (Zou et al., 2023) and Activation Addition (Turner et al., 2023) demonstrate that behavior can be modulated at inference time by adding interpretable steering vectors directly to the residual stream. Similarly, Inference-Time Intervention

(ITI) has successfully elicited truthfulness by shifting hidden activations (Li et al., 2024). While these paradigms typically target sentiment or refusal, we extend them to formal logical reasoning across multiple architectures.

Latent Knowledge and Unfaithful Reasoning. LLMs often compute correct internal representations that fail to manifest in behavioral output due to decoding biases. Burns et al. (2023) showed that linear probes can extract latent truth even when models generate incorrect text. Furthermore, Chain-of-Thought (CoT) traces (Wei et al., 2022) are frequently unfaithful to a model's true internal state (Turpin et al., 2024). We build on these findings by formalizing the **Knowledge-Behavior Gap**—a structural misalignment between latent logic and auto-regressive decoding—and providing causal evidence that activation steering can bridge this gap in both `gemma-2-2b-it` and `phi-2`.

3 Experimental Methodology

3.1 Synthetic Logic Clusters & Probing

To isolate logical reasoning from linguistic priors, we generated 250 logic clusters defined by propositional rules, a ground-truth fact, and an entailed target $\mathcal{T} \in \{\text{True}, \text{False}\}$. We randomized entity names to bypass pre-training biases and ensure a noise-free diagnostic environment, which is essential for isolating causal steering vectors without syntactic confounders. For `gemma-2-2b-it` and `phi-2`, we extracted hidden states $h_l \in \mathbb{R}^d$ at the final token position. We trained logistic regression probes $P_l(h_l) = \sigma(w_l^T h_l + b_l)$ (Alain & Bengio, 2017) using an 80/20 stratified split, downsampling the majority class to exactly 50% to eliminate label-frequency bias.

3.2 Intervention Depth & Vector Extraction

Layer-wise probing revealed an internal "saturation point" for abstract reasoning, avoiding early-layer syntax and final-layer token commitment. We identified this peak at Layer 16 ($L = 16$) for `gemma-2-2b-it` and Layer 8 ($L = 8$) for `phi-2`. To extract the causal steering direction, let $\mathcal{D}_{true}$ and $\mathcal{D}_{false}$ represent the target subsets at the saturation layer L. We define the truth vector $\mathbf{v}_{truth}$ as the difference in mean activations:

$$\mathbf{v}_{truth} = \frac{1}{|\mathcal{D}_{true}|} \sum_{i \in \mathcal{D}_{true}} h_L^{(i)} - \frac{1}{|\mathcal{D}_{false}|} \sum_{j \in \mathcal{D}_{false}} h_L^{(j)} \quad (1)$$

3.3 Activation Steering

During auto-regressive generation, we perform activation addition by modifying the residual stream at the identified saturation layers ($L = 16$ for Gemma, $L = 8$ for Phi-2): $\tilde{h}_L = h_L + \alpha \cdot \mathbf{v}_{truth}$. We sweep the coefficient $\alpha \in [0.0, 4.0]$ to identify the optimal boundary between logical recovery and linguistic degradation across both architectures.

4 Results: The Mechanistic Signature of Logic

4.1 Internal Saturation & The Solver-Augmented Paradox

At Layer 0, probing accuracy for both models is $\sim 50.0\%$, confirming no information leakage. A rapid phase transition occurs in early layers: `gemma-2-2b-it` achieves $\geq 94.0\%$ accuracy by Layer 8, while `phi-2` reaches 94.0%–98.0% saturation by Layer 8 across conditions (Figures 1 and 2). Both models achieve a full 100.0% saturation by Layer 16. Despite this near-perfect latent knowledge in the mid-layers, behavioral output plateaus (Table 1).

We identify a *Solver-Augmented Paradox*: providing explicit ground-truth symbolic logic via a Z3 solver (**Solver-Augmented**) yielded 100% internal accuracy by Layer 8 in both models, yet behavioral accuracy remained severely bottlenecked (64.0% for Gemma, 65.0% for Phi-2). This suggests a profound decoupling: even when correct logical entailment is explicitly represented in the mid-to-late residual stream, the final unembedding layer frequently overrides this signal during auto-regressive generation.

4.2 Bridging the Gap via Steering & Coherence Trade-offs

Applying the truth vector $\mathbf{v}_{truth}$ exclusively to "stubborn failures," we observe a distinct phase transition across both architectures. Baseline intervention ($\alpha = 0.0$) yields 0% recovery, whereas our optimal steering coefficient ($\alpha = 1.0$)

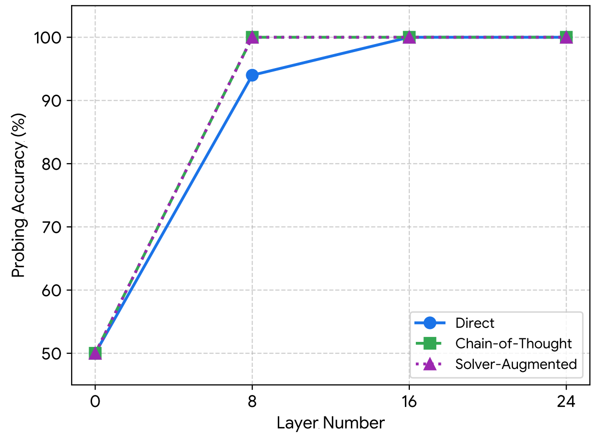

Figure 1: Layer-wise probing accuracy in `gemma-2-2b-it` for Direct, CoT, and Solver-Augmented conditions.

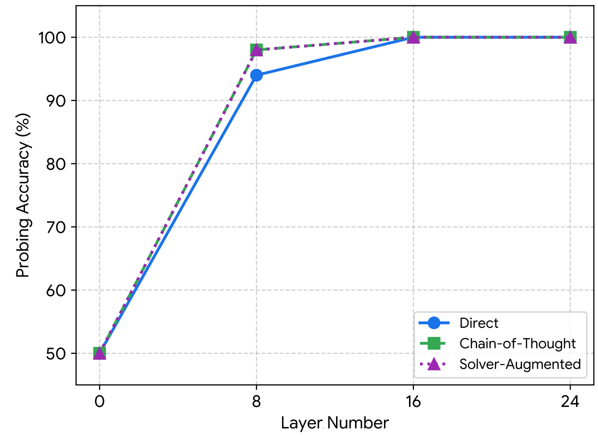

Figure 2: Layer-wise probing accuracy in `phi-2 (2.7B)` for Direct, CoT, and Solver-Augmented conditions.

Model	Condition	L0	L8	L16	L24	Behav. Acc.
Gemma-2	Direct	50.0%	94.0%	100.0%	100.0%	62.4%
	CoT	50.0%	100.0%	100.0%	100.0%	65.6%
	Solver	50.0%	100.0%	100.0%	100.0%	64.0%
Phi-2	Direct	50.0%	94.0%	100.0%	100.0%	65.0%
	CoT	50.0%	98.0%	100.0%	100.0%	65.0%
	Solver	50.0%	98.0%	100.0%	100.0%	65.0%

Table 1: Layer-wise Probing vs. Behavioral Accuracy. Latent truth saturates early across all conditions, while behavior consistently plateaus at $\sim 65\%$.

successfully recovers 76.7% of logical failures in `gemma-2-2b-it` (Figure 3) and 75.0% in `phi-2` (Figure 5). To ensure causal specificity, we conducted a control experiment by injecting random vectors of equivalent norm at the respective saturation layers; this yielded 0% recovery and accelerated linguistic degradation, confirming that $\mathbf{v}_{truth}$ targets a specific logical manifold. Beyond $\alpha = 1.5$, performance degrades in both models, suggesting interference with natural residual stream dynamics.

We evaluated the safety of this optimal $\alpha = 1.0$ intervention across two dimensions:

Robustness: Applied to 100 "Easy" queries the baseline solved perfectly, steered performance remained near-perfect (98.0% for Gemma), confirming the vector acts as a precise logical nudge rather than a blunt override.

Coherence: We quantified structural integrity via a Unique Word Ratio (UWR). At $\alpha = 1.0$, both models exhibit zero linguistic breaks, maintaining UWRs nearly identical to their baselines (~ 0.44 for Gemma and 0.382 for Phi-2). Rapid deterioration occurs only at extreme levels ($\alpha > 2.5$), indicating the underlying linguistic manifold is overridden (Figures 4 and 6).

5 Discussion and Conclusion

Evaluating `gemma-2-2b-it` and `phi-2` (2.7B) confirms that near-perfect latent logical representation ($\geq 94\%$ probing accuracy by Layer 8) is a shared characteristic of mid-scale LLMs. The *Solver-Augmented Paradox*—where behavioral accuracy plateaus at $\sim 65\%$ despite explicit ground-truth prompts and near-perfect internal saturation—demonstrates that the primary reasoning bottleneck is not representational depth, but final-layer calibration during decoding.

This mechanistically explains unfaithful Chain-of-Thought reasoning (Turpin et al., 2024): auto-regressive generation frequently overrides the correct latent conclusion in favor of linguistically probable, yet logically incorrect, output. By applying a calibrated steering vector ($\alpha = 1.0$) at saturation layers, we successfully aligned linguistic

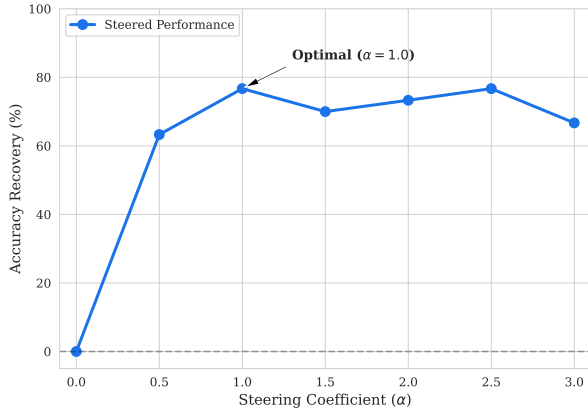

Figure 3: Steering sensitivity on stubborn failures. Layer 16 injection yields 76.7% recovery at $\alpha = 1.0$.

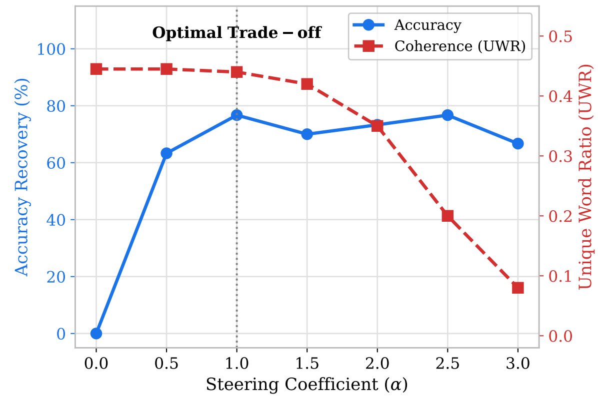

Figure 4: The coherence trade-off. At optimal recovery ($\alpha = 1.0$), UWR is stable. Degradation occurs at $\alpha > 2.5$.

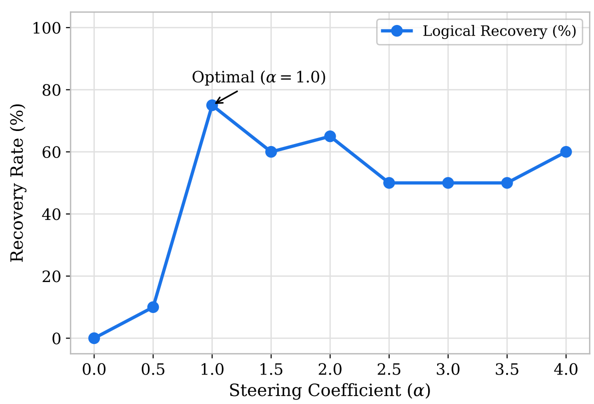

Figure 5: Steering sensitivity on stubborn failures. Layer 8 injection yields 75.0% recovery at $\alpha = 1.0$.

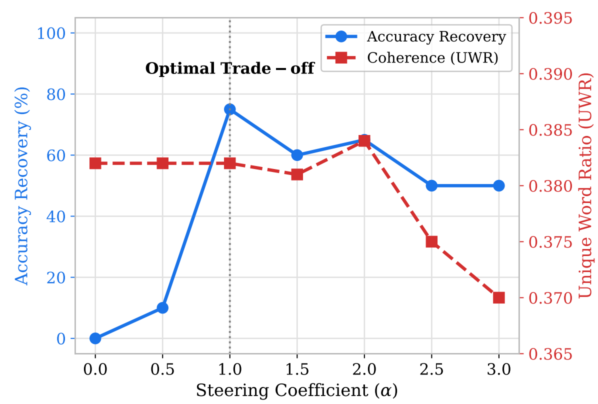

Figure 6: The coherence trade-off. At optimal recovery ($\alpha = 1.0$), UWR remains stable near 0.382. Degradation occurs at $\alpha > 2.5$.

generation with the internal state, recovering 76.7% (Gemma) and 75.0% (Phi-2) of stubborn logical failures without degrading linguistic coherence. Control experiments with random vector injections yielded 0% recovery, confirming the causal specificity of this linear truth manifold.

Limitations & Future Work: While this study validates the Knowledge-Behavior Gap across two distinct architectures, our analysis is constrained to the 2B-3B parameter regime and structurally uniform synthetic logic clusters. Future work is required to determine if this final-layer decoding bottleneck scales to frontier models and generalizes to unstructured natural language reasoning.

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